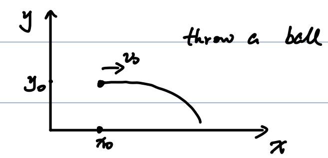
Vectors

Beyond 1-dimension.

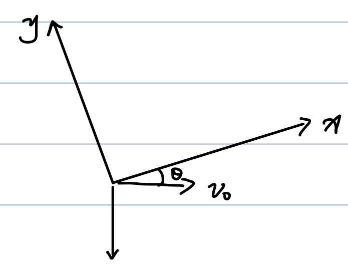


High School Way:

$$\begin{cases} x = x_0 + v_0 t \\ y = y_0 - \frac{1}{2} g t^2 \end{cases}$$

(Need track each component carefully)

Different reference frame:



$$\begin{cases} x = x_0' + v_0 \cos\theta \, t - \frac{1}{2} g \sin\theta \, t^2 \\ y = y_0' - v_0 \sin\theta \, t - \frac{1}{2} g \cos\theta \, t^2 \end{cases}$$

(highly cumbersome)

More troubles for

- higher dimensions
- many particles

A more convenient notation using vectors (矢量, 向量)

$$\begin{cases} \vec{v} = \vec{v}_0 + \vec{a} t \\ \vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2} \vec{a} t^2 \end{cases}$$

Definition of vector

- direction
- magnitude
- rules on addition

e.g.

$\vec{r}$, $\vec{v}$, $\vec{a}$ (put " $\rightarrow$ " or " $\rightharpoondown$ " in hand writing)

(bold font also commonly used in typing)

Advantages:

- concise
- independent of reference frame

Scalar: $\begin{cases} \text{magnitude } \checkmark \\ \text{direction } \times \end{cases}$

e.g. T (temperature), m (mass),

$r = |\vec{r}|$

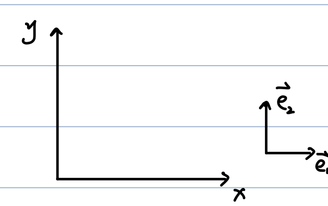
example,

position $\vec{r}$:

(choose basis)

$$\vec{r} = \{\vec{e}_1, \vec{e}_2\} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$= x\vec{e}_1 + y\vec{e}_2$$

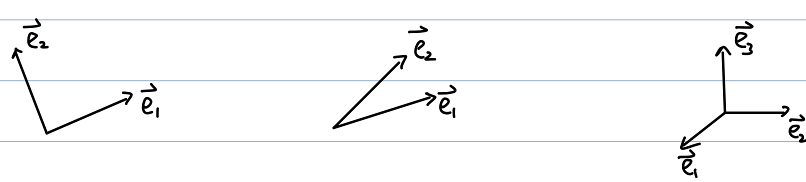


Cartesian Coordinate:

$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\vec{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\Rightarrow$ $\vec{r} = \begin{pmatrix} x \\ y \end{pmatrix}$

Generalization:



$\vec{e}_1, \vec{e}_2, \cdots, \vec{e}_n$ linearly independent

$$\vec{a} = \{\vec{e}_1, \vec{e}_2, \cdots \vec{e}_n\} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

$$\vec{b} = \{\vec{e}_1, \vec{e}_2, \cdots \vec{e}_n\} \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

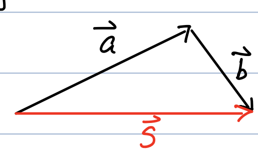
Vector addition:

$$\vec{a} + \vec{b} = \sum_i a_i \vec{e}_i + \sum_i b_i \vec{e}_i$$

$$= \sum_i (a_i + b_i) \vec{e}_i$$

$$= \{\vec{e}_1, \vec{e}_2, \cdots \vec{e}_n\} \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{pmatrix} \qquad (*) \qquad \text{(keep)}$$

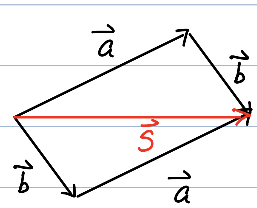
Schematically.



(head-tail)

Commutative law:

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

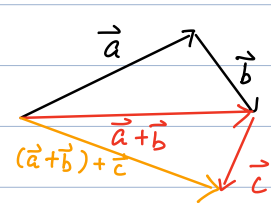


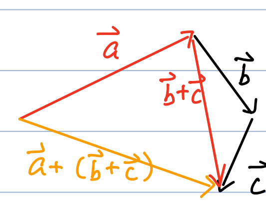
$$\{\vec{e}_1, \cdots, \vec{e}_n\} \begin{pmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{pmatrix}$$

$$= \{\vec{e}_1, \cdots \vec{e}_n\} \begin{pmatrix} b_1 + a_1 \\ \vdots \\ b_n + a_n \end{pmatrix}$$

Associative law:

$$(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$$





(In fact, just an extension of scalar addition)

$$\{\vec{e}_1, \cdots \vec{e}_n\} \begin{pmatrix} (a_1+b_1)+c_1 \\ \vdots \\ (a_n+b_n)+c_n \end{pmatrix} = \{\vec{e}_1, \cdots \vec{e}_n\} \begin{pmatrix} a_1+(b_1+c_1) \\ \vdots \\ a_n+(b_n+c_n) \end{pmatrix}$$

(Note: can not apply these laws to other objects, e.g. matrix)

$$R_1 = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix} \qquad R_2 = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$$

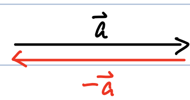
$$R_1 R_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$\neq \qquad\qquad \text{Non-commutative}$$

$$R_2 R_1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Negative:

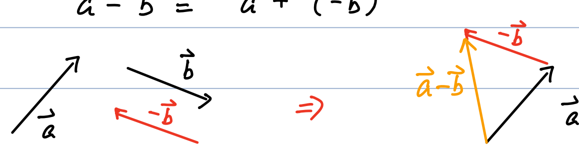
$$\vec{a} + (-\vec{a}) = 0 \qquad\qquad \sum_i a_i \vec{e}_i + (\ ? \ ) = 0$$



$$? = -\left(\sum_i a_i \vec{e}_i\right)$$
$$= \sum_i (-a_i)\vec{e}_i$$

Subtraction:

$$\vec{a} - \vec{b} \equiv \vec{a} + (-\vec{b})$$



$$\sum_i a_i \vec{e}_i - \sum_i b_i \vec{e}_i = \sum_i (a_i - b_i)\vec{e}_i$$

Multiply by scalar:

$$\alpha \vec{a} = \alpha \sum_i a_i \vec{e}_i$$
$$= \sum_i (\alpha a_i)\vec{e}_i$$

(component-by-component)

Scalar product of vectors:

$$\vec{a}\cdot\vec{b} = \Big(\sum_i a_i \vec{e}_i\Big)\cdot\Big(\sum_j b_j \vec{e}_j\Big)$$
$$= \sum_i\sum_j a_i b_j \; \vec{e}_i\cdot\vec{e}_j$$

For cartesian coordinate:

$$\vec{e}_i\cdot\vec{e}_j = \delta_{ij} \qquad \left(\begin{array}{l} \delta_{ij}=1 \quad \text{for} \quad i=j \\ \delta_{ij}=0 \quad \text{for} \quad i\neq j \end{array}\right)$$

$$\Rightarrow \quad \vec{a}\cdot\vec{b} = \sum_i\sum_j a_i b_j \delta_{ij} \qquad (\text{cartesian})$$
$$= \sum_i a_i \Big(\sum_j b_j \delta_{ij}\Big)$$
$$= \sum_i a_i b_i$$
$$\equiv (a_1 \;\; a_2 \;\cdots\; a_n)\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

Note 1 : Commutative law:

$$\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{a}$$

Note 2 : For non-cartesian coordinates, often

$\vec{a}\cdot\vec{b}$ can NOT be written as $(a_1 \cdots a_n)\cdot\begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$

Length:

$$r \equiv |\vec{r}| = \sqrt{\vec{r}\cdot\vec{r}}$$

Unit vectors: (length 1)

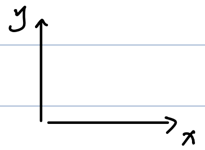
$\hat{i}, \hat{j}, \hat{k},$

$(\hat{x}, \hat{y}, \hat{z})$

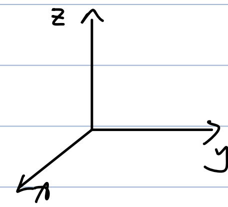
$(\hat{e}_x, \hat{e}_y, \hat{e}_z)$

More on Cartesian coordinates:

2D



3D



$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\quad \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ $\qquad \vec{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $\quad \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ $\quad \vec{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$\hat{i}$ $\quad \hat{j}$ $\qquad \hat{i}$ $\quad \hat{j}$ $\quad \hat{k}$

$\hat{x}$ $\quad \hat{y}$ $\qquad \hat{x}$ $\quad \hat{y}$ $\quad \hat{z}$

$\hat{e}_x$ $\quad \hat{e}_y$ $\qquad \hat{e}_x$ $\quad \hat{e}_y$ $\quad \hat{e}_z$

We already know:

$$\vec{a} \cdot \vec{b} = \sum_i a_i b_i \qquad \text{(cartesian)}$$

$$\Rightarrow \quad \vec{a} \cdot \vec{a} = \sum_i (a_i)^2$$

$$\Rightarrow a = |\vec{a}| = \sqrt{\sum_i (a_i)^2} = \begin{cases} \sqrt{x^2 + y^2} & (2D) \\ \sqrt{x^2 + y^2 + z^2} & (3D) \end{cases}$$

More general definition on unit vector $\hat{r}$:

$$|\hat{r}| = 1$$

Any vector:

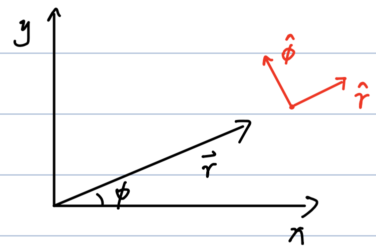
$$\vec{r} = r\,\hat{r}$$

e.g. $\quad \vec{r} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \Rightarrow \quad r = 5\,\hat{r}$

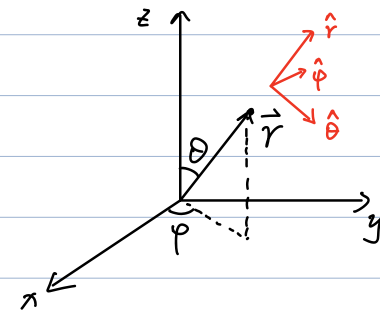
$$\hat{r} = \begin{pmatrix} \frac{3}{5} \\ \frac{4}{5} \end{pmatrix}$$

Polar coordinates:

2D



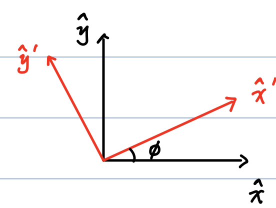
3D



$$\begin{cases} x = r\cos\phi \\ y = r\sin\phi \end{cases}$$

$$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases}$$

Basis transformation:



$$\begin{cases} \hat{x}' = \cos\phi\,\hat{x} + \sin\phi\,\hat{y} \\ \hat{y}' = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y} \end{cases}$$

$$\{\hat{x}', \hat{y}'\} = \{\hat{x}, \hat{y}\}\begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}$$

$$\vec{r} = \{\hat{x}, \hat{y}\}\begin{pmatrix} x \\ y \end{pmatrix}$$

$$= \{\hat{x}', \hat{y}'\}\begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\equiv \{\hat{x}', \hat{y}'\}\begin{pmatrix} x' \\ y' \end{pmatrix}$$

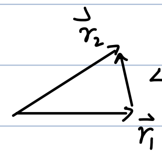
$$\Rightarrow \begin{cases} x' = \cos\phi\, x + \sin\phi\, y \\ y' = -\sin\phi\, x + \cos\phi\, y \end{cases}$$

Back to Physics (1D → 2D, 3D)

(position no longer on a straight line)

position: $\vec{r}$

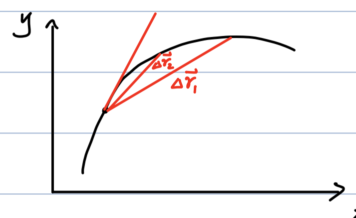
displacement: $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$



$\Delta\vec{r} \not\propto \vec{r}_1$ or $\vec{r}_2$

average velocity: $\vec{v}_{avg} = \dfrac{\Delta\vec{r}}{\Delta t}$ $\quad\left[\ \left(\dfrac{1}{\Delta t}\right)\Delta\vec{r}\ \right]$

instantaneous velocity: $\vec{v} = \lim\limits_{\Delta t\to 0} \dfrac{\Delta\vec{r}}{\Delta t} = \dfrac{d\vec{r}}{dt}$



($\vec{v}$ tangent to path)

average acceleration: $\vec{a}_{avg} = \dfrac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1}$

instantaneous acceleration: $\vec{a} = \lim\limits_{\Delta t\to 0} \dfrac{\Delta\vec{v}}{\Delta t} = \dfrac{d\vec{v}}{dt}$

By componement:

$$\vec{r} = x\,\hat{i} + y\,\hat{j} + z\,\hat{k}$$

$$\Rightarrow \vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$$

$$= v_x\,\hat{i} + v_y\,\hat{j} + v_z\,\hat{k}$$

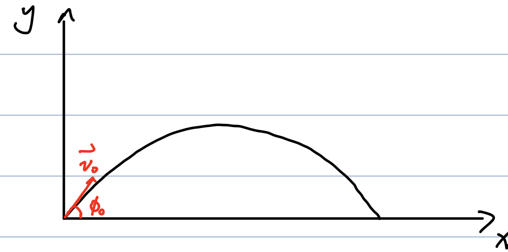
Similarly,

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} + \frac{dv_z}{dt}\hat{k}$$

$$= \frac{d^2x}{dt^2}\hat{i} + \frac{d^2y}{dt^2}\hat{j} + \frac{d^2z}{dt^2}\hat{k}$$

$$= a_x\,\hat{i} + a_y\,\hat{j} + a_z\,\hat{k}$$

Projectile motion. (constant acceleration $\vec{a} = -g\,\hat{j}$)



$$\vec{r} = \underbrace{\vec{r}_0}_{0} + \vec{v}_0 t + \frac{1}{2}\underbrace{\vec{a}}_{-g\hat{j}} t^2$$

$\vec{r}_f = s\,\hat{i} + 0\,\hat{j}$ (most convenient basis)

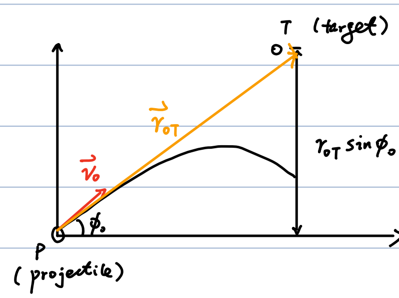
$$\Rightarrow \begin{pmatrix} s \\ 0 \end{pmatrix} = \begin{pmatrix} v_0 \cos\phi_0 t \\ v_0 \sin\phi_0 t \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 0 \\ -gt^2 \end{pmatrix}$$

$$\Rightarrow \quad t = \frac{2 v_0 \sin\phi_0}{g}$$

$$s = \frac{v_0^2 \sin 2\phi_0}{g}$$

(choosing a proper basis greatly simplifies the calculation)

Shooting a falling object :



Q1 : where to aim ?

(frame change : $-g\,\hat{j}$) $\rightarrow$ (uniform speed $\vec{v}_0$)

Q_2 : minimal speed ?

$$\vec{r}_{oT} = \vec{v}_0 t$$

$$\left.\begin{aligned} &\Rightarrow \quad t = \frac{r_{oT}}{v_{oT}} \\ &\text{critical:} \quad r_{oT}\sin\phi_0 = \frac{1}{2} g t^2 \end{aligned}\right\} \Rightarrow \quad \frac{2 r_{oT}\sin\phi_0}{g} = \left(\frac{r_{oT}}{v_{oT}}\right)^2$$

$$\Rightarrow \quad v_{oT} = \sqrt{\frac{g\, r_{oT}}{2\sin\phi_0}}$$

(Actually, no need to change frame)

$$\vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a} t^2$$

$$\begin{cases} \vec{r}_{op} = \vec{v}_0 t - \frac{1}{2} g t^2 \hat{j} \\ \vec{r}_{oT} = (r_{oT}\cos\phi_0 \hat{i} + r_{oT}\sin\phi_0 \hat{j}) - \frac{1}{2} g t^2 \hat{j} \end{cases}$$

(Use $\hat{i}, \hat{j}$ frame)

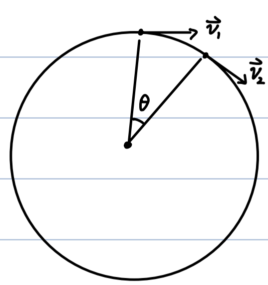
$$\vec{r}_{op} = \vec{r}_{oT}$$

$$\Rightarrow \quad \vec{v}_0 t = r_{oT}\cos\phi_0 \hat{i} + r_{oT}\sin\phi_0 \hat{j}$$

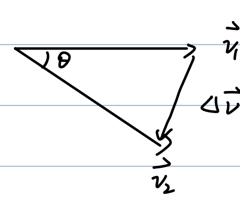
$$\Rightarrow \quad t \begin{pmatrix} v_{oT}\cos\phi \\ v_{oT}\sin\phi \end{pmatrix} = r_{oT} \begin{pmatrix} \cos\phi_0 \\ \sin\phi_0 \end{pmatrix}$$

$$\Rightarrow \quad \begin{cases} \phi = \phi_0 \\ t = r_{oT} / v_{oT} \end{cases}$$

Uniform circular motion



$$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta \vec{v}}{\Delta t} \quad \text{points to center}$$



$$\theta = v\Delta t / r$$

$$|\Delta \vec{v}| = 2v \sin\frac{\theta}{2}$$

$$\Rightarrow \quad |\vec{a}| = \lim_{\Delta t \to 0} \frac{2v\sin\frac{\theta}{2}}{\Delta t}$$

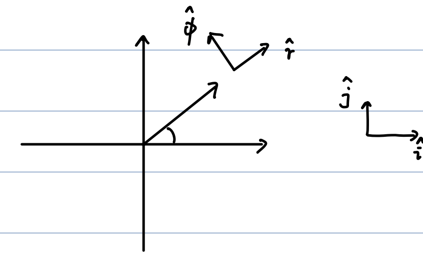
$$= \lim_{\Delta t \to 0} \frac{2v \sin\frac{v\Delta t}{2r}}{\Delta t}$$

$$= \frac{v^2}{r} \quad \Rightarrow \quad \vec{a} = -\frac{v^2}{r}\hat{r} \quad \text{(centripetal acceleration)}$$

Formally,

Using polar coordinates:

$$\{\hat{r}, \hat{\phi}\} = \{\hat{i}, \hat{j}\} \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}$$



$$\Rightarrow \frac{d\hat{r}}{dt} = \frac{d\cos\phi}{dt}\hat{i} + \frac{d\sin\phi}{dt}\hat{j}$$

$$= -\sin\phi \frac{d\phi}{dt}\hat{i} + \cos\phi\frac{d\phi}{dt}\hat{j}$$

$$= -\sin\phi\frac{d\phi}{dt}(\cos\phi\,\hat{r} - \sin\phi\,\hat{\phi})$$

$$+ \cos\phi\frac{d\phi}{dt}(\sin\phi\,\hat{r} + \cos\phi\,\hat{\phi})$$

$$= \frac{d\phi}{dt}\hat{\phi}$$

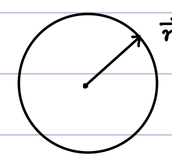
$$\{\hat{i}, \hat{j}\} = \{\hat{r}, \hat{\phi}\}\begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix}$$

Similarly,

$$\frac{d\hat{\phi}}{dt} = -\frac{d\phi}{dt}\hat{r}$$

Uniform circular motion: $\frac{d\phi}{dt} = w = \frac{v}{r}$

$\vec{r} = r\hat{r}$ ($\hat{r}$ is a function of t)



$$\vec{v} = \frac{d\vec{r}}{dt} = r\frac{d\hat{r}}{dt}$$

$$= r\frac{d\phi}{dt}\hat{\phi}$$

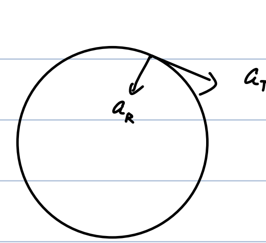
$$= wr\hat{\phi}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = wr\frac{d\hat{\phi}}{dt}$$

$$= wr\left(-\frac{d\phi}{dt}\hat{r}\right)$$

$$= -w^2 r\hat{r} = -\frac{v^2}{r}\hat{r}$$

Note: non-uniform circular motion



$\vec{v} = \frac{d\vec{r}}{dt} = wr\hat{\phi}$ (same) $\quad w = \frac{d\phi}{dt}$ (time-dependent)

$$\vec{a} = \frac{d\vec{v}}{dt} = \underbrace{\frac{dw}{dt}r\hat{\phi}}_{\text{tangential}} + \underbrace{wr\frac{d\hat{\phi}}{dt}}_{\text{centripetal}}$$